01−Teensy−AudioShield

File: sheets/01−teensy.kicad_sch

02−Analog−Front

File: sheets/02−analog−front.kicad_sch

03−ADC−ADS1115

File: sheets/03−adc.kicad_sch

04−MCP23017−IO

File: sheets/04−mcp−io.kicad_sch

05−CV−DAC

File: sheets/05−cv−dac.kicad_sch

06−Surface−Inputs

File: sheets/06−surface.kicad_sch

07−MIDI−DIN

File: sheets/07−midi.kicad_sch

08−Connectors−Panel

File: sheets/08−connectors−panel.kicad_sch

09−Connectors−Rear

File: sheets/09−connectors−rear.kicad_sch

10−Power

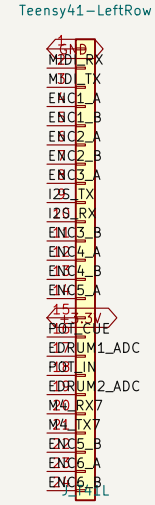
File: sheets/10−power.kicad_sch

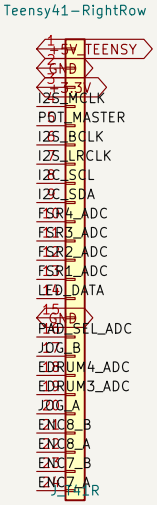
Sheet: /
File: pcb−m.kicad_sch

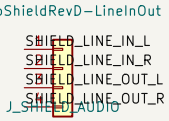
Title:

Size: A4 | Date: | Rev:

KiCad E.D.A. 10.0.6 | Id: 1/11









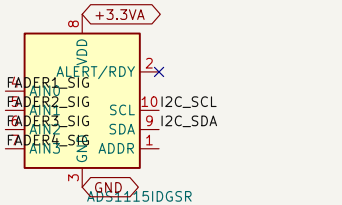
NOTAS TEENSY 4.1 (D2):\nMódulo Audio Shield Rev D en zócalos J_T41L/J_T41R (montaje tipo shield lado a lado, no apilado)\nSGTL5000 en el shield — I2S 7/8/20/21/23 y I2C 18/19 pasan por el socket\nCortar puente VUSB–VIN en la Teensy; alimentar desde USB-A del Pi (separado del rail de LEDs)\nSin microSD SPI (pines 10/11/12/13 son super

Sheet: /01–Teensy–AudioShield/		
File: 01–teensy.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/11

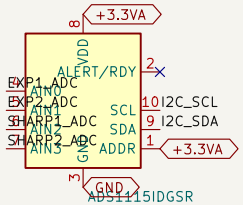
Sheet: /02-Analog-Front/
File: 02-analog-front.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/11



ADS1115 #1 (0x48) — Faders 1–4



ADS1115 #2 (0x49) — EXP1/2 + Sharp1/2

Sheet: /03–ADC–ADS1115/
File: 03–adc.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 4/11

MCP23017 #4 (0x23) PCB-M\nMotor en vivo + jog + pedal\nGPA7/GPB7 output-only errata: NC

Sheet: /04-MCP23017-IO/
File: 04-mcp-io.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 5/11

VDD
MCP4725A0T-E/CH
GND

CV DAC — MCP4725 x2 + MCP6004 buffer gain 1.5\nU_DAC1 (0x60): A0=GND; U_DAC2 (0x61): A0=+3.3V\nBuffer gain: Rf=10k Rin=20k (1+10k/20k=1.5)\nSalida: 0–3.3V*1.5=0–4.95V aprox. 5V\nSerie 1kohm + BAT54S a GND/+5V_LED por salida\nT

Sheet: /05-CV-DAC/
File: 05-cv-dac.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/11



JOG: EC11 Ø30–40mm con clic\nA→T37, B→T40, SW→MCP23/GPA6\nFSR velocity: 10kΩ pulldown a GND\nT14–T17=FSR1–4_ADC (ADC1 2.5kHz ISR)\nPots: RC 1kΩ/100nF post–wiper\nPOT_MASTER→T22(ADC2), POT_CUE→T24, POT_IN→T26\nTOD

Sheet: /06–Surface–Inputs/ File: 06–surface.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/11

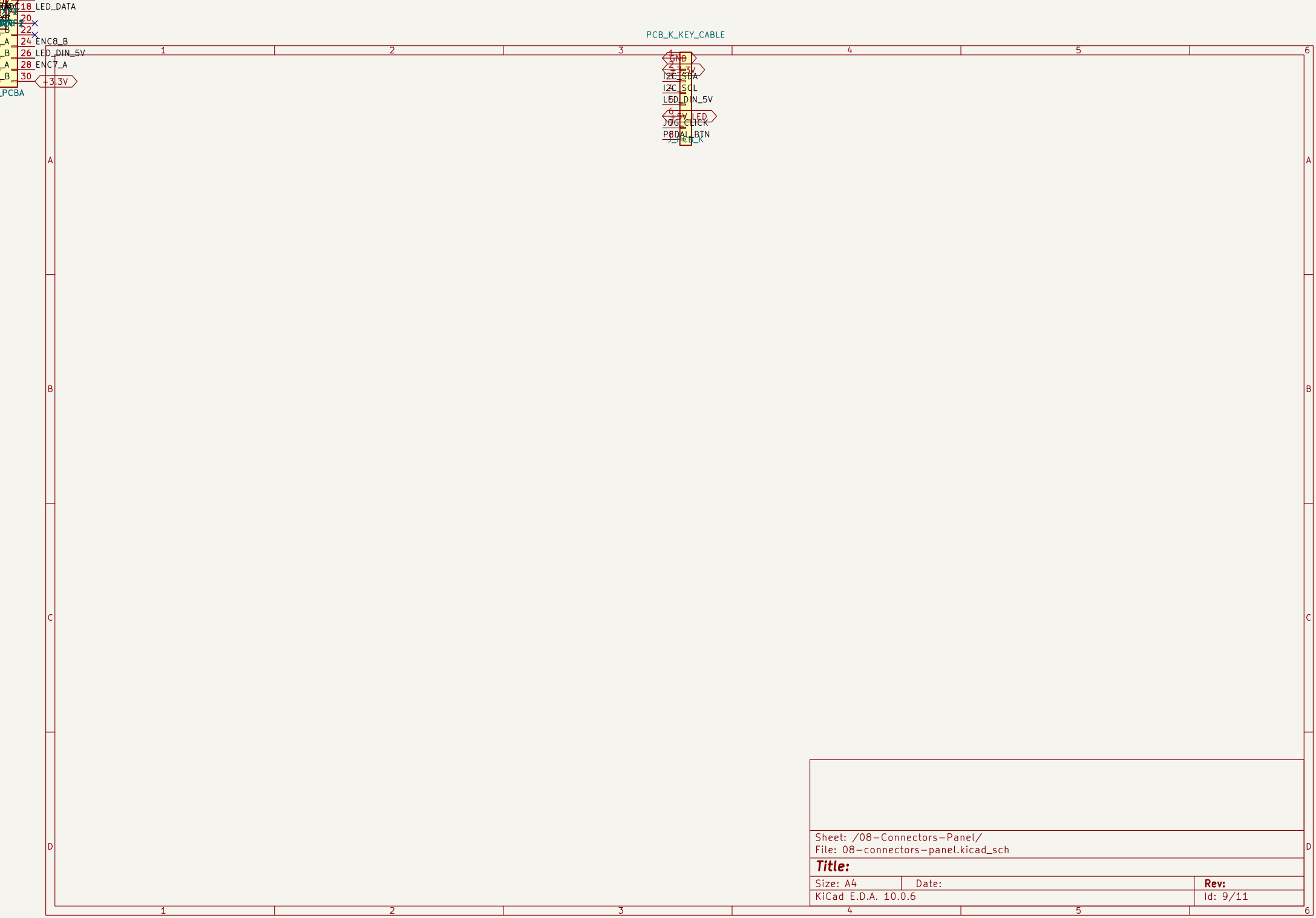


MIDI TRS Tipo A (MIDI Assoc. spec)\nIN: T=GND (100ohm), R=anode 6N138 (220ohm serie), S=GND\n6N138: VO1+VO2=MIDI_RX (T0/Serial1), 220ohm pull-up a 3.3V\nOUT: T+R=MIDI_TX (T1/Serial1) a traves de 47ohm c

Sheet: /07-MIDI-DIN/
File: 07-midi.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/11



Sheet: /08-Connectors-Panel/
File: 08-connectors-panel.kicad_sch

Title:

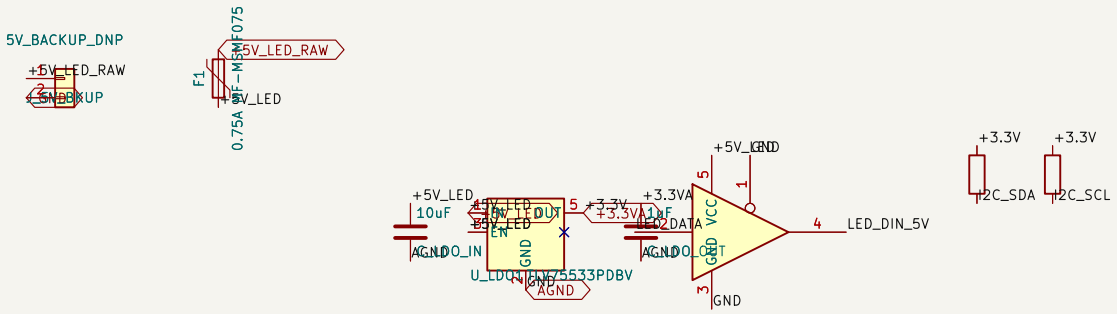
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/11



CONECTORES TRASEROS Y FRENTE
LINE IN: NRJ3HF-1 6.35mm TRS | LINE OUT: 6.35mm TS mono buffered | HP/CUE: 6.35mm TRS
CV1/CV2: 3.5mm TS (0-5V) | PEDAL: 6.35mm TRS (Tip=PEDAL_BTN → MCP23017 GPB0, GND)
J_EDRUM: DNP/reservado (5 pines para triggers e-drum futura expansion)

Sheet: /09-Connectors-Rear/		
File: 09-connectors-rear.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 10/11

EN tied a IN (always enabled). 100nF + 1uF decoupling en IN y OUT.
U_LED_BUF: 74AHCT1G125 — level shifter LED_DATA (3.3V Teensy T13) → LED_DIN_5V (5V)
OE (pin1) = GND (siempre habilitado). VCC = +5V_LED.
R_SDA/R_SCL: 2.2kΩ pull-ups I2C. NOTA: remover R1/R2 de PCB-A cuando PCB-M instalado.
+5V_LED rail: desde J1_PCBA pin 28 (USB-A polyfused). NO conectar a +5V digital.



NOTAS DE ALIMENTACIÓN (D6):\n+5V_TEENSY: desde USB-A Pi → USB micro-B Teensy (cortar puente VUSB-VIN en Teensy)\n+5V_LED_RAW: desde USB-A Pi (3.° puerto) → J_5V_BKUP (plan B sin poblar) + F1 (polyfuse 0.75A)\n+5V_LED: post-polyfuse → LEDs SK6812, Sharp GP2Y0A41\n+3.3VA: TLV75533 regulado desde +5V_LED → opamps, ADS, MCP2301

Sheet: /10-Power/		
File: 10-power.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 11/11